

Assignment 1 Dot Net Programming & Dot Net Basics

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Q.1. What is the use of programming language? Write its brief history.

→ A programming language is used to write instructions that tell a computer what to do. It helps in developing software, websites, mobile apps, games & operating system.

→ Brief history:-

- 1940s : machine language (0s and 1s)
- 1950s : Assembly Language
- 1957 : FORTRAN introduced for scientific computing.
- 1959 : COBOL introduced for business applications.
- 1972 : C Language developed by Dennis Ritchie.
- 1980s - 1990s : C++, Java, and Visual Basic became popular.
- 2000 : C# was introduced by Microsoft with the .NET Framework.
- Today, languages like Python, JavaScript, C# and Java are widely used.

Q.2 Why is .NET Technology used for creating enterprise Applications? Explain the Enterprise Gap and how .NET reduces it.

→ .NET is used for Enterprise Applications are:-

- Easy to develop large applications.
- High security
- Fast performance
- Supports multiple programming languages
- Easy database connectivity.
- Cross-platform support with .NET.

• Enterprise Gap:- Enterprise Gap is the difference betⁿ the growing needs of businesses & the availability of modern software solutions. In Nepal, many organizations still use manual or outdated systems due to limited resources & skilled developers.

• AJF reduces this gap:-

- Helps build secure & reliable applications.
- Supports web, desktop, & cloud applications.
- Reduces development time.
- Easy maintenance & scalability.
- Integrates easily with databases & business systems.

Q.3. What are the uses of C# programming? What's new in C#?

→ Uses of C#:-

- Windows desktop applications.
- Web applications using ASP.NET.
- Mobile applications.
- Game development using Unity.
- Cloud applications.
- Enterprise software.

→ What's new in C# are:-

- Better performance
- Nullable Reference Types
- Records for simpler data objects
- Pattern matching
- Top-level Statements
- Improved async programming
- Cross-platform development with modern .NET.

Q. 4. What is CLR Technology? Explain CLS, CTS, MSIL and JIT.

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- CLR (Common Language Runtime) :- CLR is the execution engine of the .NET framework. It manages memory, security, exception handling, & executes .NET programs.
- CLS (Common Language Specification) :- A set of rules that all .NET languages follow so they can work together.
- CTS (Common Type System) :- Defines common data types & programming rules used by all .NET languages.
- MSIL (Microsoft Intermediate Language) :- The intermediate code generated when C# source code is compiled.
- JIT (Just-In-Time Compiler) :- Converts MSIL into machine code just before execution so the CPU can run it.

Q. 5. What is Visual Studio? Why is it popular?

- Visual Studio is an Integrated Development Environment (IDE) developed by Microsoft for creating desktop, web, mobile, & cloud applications.

→ It is popular because of :-

- easy to use interface
- powerful code editor

- Debugging tools

- IntelliSense (code suggestions).

- Drag & drop GUI designer

- Git integration

- Supports many programming languages like C#, C++, Python, & Javascript.

- fast application development with .NET.

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